

Week 1: Python Programming Fundamentals

Topics: Introduction to Python, Applications of Python, Python Installation, VS Code Setup, Running Python Programs, Variables, Data Types, User Input & Output, Type Casting, Arithmetic Operators, Assignment Operators, Comparison Operators, Logical Operators, Membership Operators, Identity Operators, Truthy and Falsy Values, Conditional Statements (if, if-else, if-elif-else, Nested if, Ternary Operator), Loops (while, for, break, continue, pass, Nested Loops), Strings (Indexing, Slicing, String Methods, String Formatting, f-Strings), Collections (List, Tuple, Set, Dictionary), Functions, Arguments, Return Values, Lambda Functions, Modules, Packages, OOPS

Mini Project: Student Management System

Week 2: Data Analysis & Visualization

Topics: NumPy Arrays, Dimensions, Indexing, Slicing, Reshaping, Broadcasting, Mathematical Operations, Statistical Operations, Random Module, Pandas Series, DataFrames, Reading CSV Files, Writing CSV Files, Missing Values Handling, Duplicate Values Handling, Data Filtering, Sorting, GroupBy, Pivot Tables, Merge, Join, Matplotlib (Line Chart, Bar Chart, Histogram, Scatter Plot, Pie Chart), Seaborn (Count Plot, Box Plot, Heatmap, Pair Plot, Correlation Analysis)

Week 3: Machine Learning

Topics: AI vs ML vs DL, Types of Machine Learning, Machine Learning Workflow, Train-Test Split, Label Encoding, One-Hot Encoding, Feature Scaling, Standardization, Normalization, Linear Regression, Multiple Linear Regression, Logistic Regression, K-Nearest Neighbors (KNN),

Decision Tree, Random Forest, Bagging, Boosting, Gradient Boosting, Accuracy, Precision, Recall, F1 Score, Confusion Matrix, Cross Validation

Mini Project Using ML

Week 4: Generative AI, LangChain & RAG

Topics: Introduction to Generative AI, Large Language Models (LLMs), OpenAI Models, Gemini Models, Prompt Engineering, LangChain Architecture, Embeddings, Text Chunking, Similarity Search, Retrieval-Augmented Generation (RAG), RAG Architecture, Document Processing, Retrieval Pipeline, Streamlit Basics, Streamlit Components, Forms

Week 5: Capstone Project